

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-10	Team No-12	Project Title- The Society/Apartment Management Portal
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1. Abstract

The Society/Apartment Management Portal is a web-based application designed to simplify and digitize the day-to-day management activities of residential apartments and housing societies. In many apartments, activities such as maintenance fee collection, complaint handling, visitor entry, amenity booking, and communication with residents are managed manually or through separate systems. This can result in inefficient record management, delayed communication, and difficulty in tracking payments and requests.

The proposed system provides a centralized platform where apartment administrators and residents can efficiently manage these activities. The portal maintains resident and flat information and allows administrators to generate and track maintenance bills. Residents can view their maintenance dues and make online payments through the integrated payment facility. The system also supports visitor gate-pass management for maintaining visitor records and improving entry management.

The system will use a structured database to maintain resident, payment, complaint, visitor, and booking information. Overall, the project aims to provide an organized, convenient, and centralized solution for managing residential society operations.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090085	V. Sujit	
2520090087	D. Bala Charan	
2520090023	M. Sai Chaitanya	

3. Frameworks / Platforms / Software Used

Frontend: React.js (or simple HTML/CSS/JavaScript)

Backend Framework: FastAPI (Python)

Database: MongoDB

Database Driver / ODM: Motor (Async) or PyMongo / Beanie

API Testing & Tools: Postman, Swagger UI (built directly into FastAPI), Git, VS Code

Guide Name and Signature (remarks if any) _____